Harsh Gupta

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Profile

IIT Gandhinagar PGD-AI/ML candidate (2026) with 3 years of production engineering experience. Specialized in building and deploying multi-agent LLM systems using LangGraph and LangChain, with hands-on expertise in RAG pipeline design, RAG evaluation (RAGAS), vector search, and agent orchestration. Backend depth in Python, FastAPI, and cloud-native deployment used to ship AI systems to production — not just prototypes.

Education

IIT Gandhinagar, PGD-AI/ML — Feb 2026 – July 2026

- CGPA: 8.0/10.0

AKTU, B.Tech in Computer Science — July 2019 – July 2023

- CGPA: 7.6/10.0

Experience

Software Engineer, National e-Governance Division (NeGD) – New Delhi, India — Oct 2025 – Feb 2026

- Architected a **schema-based form engine** using **React + NestJS**, allowing MoSPI teams to update complex workflows across ministries and states without frontend code changes.
- Replaced hardcoded frontend forms with a **backend-driven schema architecture**, reducing change implementation time from **2–3 days to under 2 minutes** through an admin dashboard.
- Reduced development and QA cycles by enabling **real-time field creation, updates, and removal** without code deployments, improving stakeholder turnaround and delivery speed.

Software Engineer, Plutos ONE – Noida, UP — Jan 2024 – Oct 2025

- Designed and scaled backend systems for **BBPS bill fetch, payment, and confirmation workflows** using **NestJS**, processing over **3M+ transactions per month** with high availability and low-latency APIs.
- Engineered high-throughput async processing pipelines using **BullMQ + Redis**, improving execution latency by **40%** while enabling retries, failure recovery, and distributed job orchestration.
- Led end-to-end onboarding of **10+ billers** with full **NPCI/BBPS compliance**, designing reusable integration modules that enabled new biller onboarding with **<10% code changes**.
- Automated settlement and reconciliation workflows between **banks, billers, and PSPs**; improved large-file processing throughput by **60%** using **streaming pipelines, cloud storage, and optimized batch processing**.
- Implemented a production-grade microservices deployed on **Docker + Kubernetes + GCP**, with centralized logging, monitoring, and scalable cloud-native architecture for mission-critical payment systems.
- Developed an automated biller onboarding platform selected under **NPCI IdeaSpark Innovation Program**, currently scaling integrations across the BBPS ecosystem.

Software Engineer, Echo IT Solutions – Muzaffarnagar, UP — Apr 2022 – Oct 2023

- Architected a new student records module within the ERP system, streamlining data access for 50+ administrative staff and reducing processing time by 40% for transcript requests.
- Optimized reporting and transaction modules through **SQL indexing and query tuning**, improving query performance by **45%** and enabling faster analytics dashboards.
- Built maintainable API services and backend architecture supporting authentication, onboarding, transactional workflows, and operational automation across multiple products.

Technical Skills

Languages: Python, TypeScript

AI / LLM Engineering: LangChain, LlamaIndex, Vector Databases, OpenAI API, Hugging Face Transformers, RAGAS, LangSmith, Guardrails

Multi-Agent Systems: LangGraph, CrewAI, AutoGen-style Frameworks, Tool Calling, Workflow Orchestration, Agent Memory Design

ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Backend / APIs: NestJS, FastAPI, REST APIs, Microservices, Event Queues, WebSockets

Databases & Storage: PostgreSQL, MongoDB, Redis, Vector Databases

MLOps / DevOps: Docker, Kubernetes, MLflow, DVC, CI/CD (GitHub Actions, Jenkins)

Cloud Platforms: AWS (S3, EC2, Lambda), GCP

Tools: Git/GitHub, Linux, Postman, Jira, VS Code

Problem Solving: Solved **500+ LeetCode** Problems (DSA, Algorithms, System Design)

Projects

TenderLens AI | AI-Powered Tender Evaluation Platform | Github

- Developed an AI procurement platform that extracts **tender criteria**, validates submissions, and generates **eligibility + compliance reports**, reducing manual review by **80%+**.
- Engineered a **LangGraph multi-agent workflow** for parsing, evidence matching, and comparative scoring across **100+ page tender documents**.
- Improved retrieval and reasoning quality through **LangSmith + RAGAS evaluation**, increasing response grounding and pipeline reliability.
- **Tech Stack:** LangGraph, FastAPI, ReactJS, Groq, OpenRouter, Celery, Redis, LangSmith, RAGAS

IntelliGrad | Gamified AI Learning Platform | Github

- Created an AI learning platform that converts **LLM interactions into measurable scores, rewards, and progress metrics** for verifiable learning outcomes.
- Implemented **LangChain + Vector DB + Guardrails** pipelines, maintaining **95%+ topic adherence** while improving conversational relevance.
- Added **evaluation workflows** for response quality and scoring consistency, improving topic-specific accuracy across learning sessions.
- **Tech Stack:** LangChain, FastAPI, Vector DB, Guardrails, LangSmith, ReactJS

AdmitGuard | AI Admission Eligibility & Verification Platform | Github

- Designed an AI screening platform that automates **eligibility validation, exception tracking, and document verification** for admissions across premier institutions.
- Transformed spreadsheet-based verification into a **rule engine + AI workflow**, reducing manual verification cycles by **70%**.
- Introduced **evaluation pipelines** to improve rule accuracy, document validation consistency, and auditability.
- **Tech Stack:** FastAPI, ReactJS, Rule Engine, AI Validation, PostgreSQL